Final Project Report for Bachelor in Computer Engineering

HOUSE RENTAL SYSTEM

Submitted by:

Anisha Gaire	**BEC [2081-04]**
Prakriti Sharma	**BEC [2081-16]**
Purnima Poudel	**BEC [2081-17]**
Sneha Sah	**BEC [2081-26]**

United Technical College

Faculty of Science and Technology

Affiliated to Pokhara University, Nepal

July 5, 2025

Minor Project Report for the Degree of Bachelor of Computer Engineering

HOUSE RENTAL SYSTEM

Supervised By: Er. Manoj Sapkota

A third-year project report submitted in partial fulfillment of the requirements for the degree of Bachelor of Computer Engineering.

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DEDICATION

This project is dedicated to our families, whose unwavering support and encouragement have been the foundation of our efforts. We also dedicate this House Rental System to all property owners, tenants, and real estate managers who strive to bring structure and transparency to the housing market. To our mentors and educators, whose guidance and wisdom have shaped our understanding, thank you for inspiring us to build solutions that matter.

DECLARATION

We hereby affirm that this research study, titled "House Rental System" represents the culmination of our original work. Any relevant prior research conducted by other scholars has been duly acknowledged.

We assume full responsibility for the precision and authenticity of all data and information contained within this study.

Names of the Students:

Anisha Gaire

Prakriti Sharma

Purnima Poudel

Sneha Sah

Date: July 5, 2025

RECOMMENDATION

This is to certify that the project work titled “House Rental System” prepared and submitted by Anisha Gaire, Prakriti Sharma, Purnima Poudel, and Sneha Sah in partial fulfillment of the requirements for the degree of Bachelor of Engineering (BE) in Computer Science, awarded by Pokhara University, has been conducted under the supervision of Er. Manoj Sapkota.

We recommend the acceptance of this project by Pokhara University based on its comprehensive research, innovative approach, and successful execution.

Signature: _____

Name of the Supervisor: Er. Manoj Sapkota

Organization: United Technical College

Date: July, 2025

CERTIFICATE

This project entitled "**House Rental System**" was prepared and submitted by Anisha Gaire, Prakriti Sharma, Purnima Poudel, and Sneha Sah have been examined by us and are accepted for the award of the degree of Bachelor Engineering in Computer by Pokhara University.

Signature: _____

Name of External:

Designation:

Organization:

Signature: _____

Name of Supervisor: Er. Manoj Sapkota

Organization: United Technical College

Signature: _____

Name of Supervisor: Er. Sahit Baral

Organization: United Technical College

Signature: _____

Name of Principal: Prof. Dr Madhav Prasad Koirala

Organization: United Technical College

Date: July, 2025

ACKNOWLEDGEMENT

We would like to express our appreciation to all who have supported us during the completion of this project. Without their help and encouragement, it would not have been possible to complete the project.

We are highly indebted to our Supervisor **Er. Manoj Sapkota** for all his valuable, guidance, motivation, and support us during the completion of this project. His expertise and feedback were essential in shaping our ideas and improving the quality of our work.

We would like to thank our Head of Department **Er. Sahit Baral** for every support and motivation for the completion of the project. Their support and guidance played a significant role in achieving our goals.

Lastly, we would like to thank our friends and classmates for their valuable feedback and In conclusion, we are grateful to everyone who has contributed to the successful completion of this project. Their support and guidance have been vital to our academic journey, and we express our sincere gratitude to them.

ABSTRACT

The process of finding and renting houses often poses challenges, particularly for individuals seeking affordable and suitable accommodation in specific locations. These difficulties include limited property visibility, lack of authentication, and inefficient communication between tenants and landlords. To address these issues, this project introduces a House Rental System that simplifies and digitizes the house rental process.

The system allows users to register, create personalized profiles, list or browse rental properties, and manage booking requests. It supports both landlord and tenant interactions, incorporating features such as authentication, authorization, and AI-based recommendation algorithms to improve property visibility and user experience. By offering a streamlined, transparent, and secure rental process, the system benefits both tenants and landlords, making house renting more accessible and efficient.

To enhance location awareness, the system integrates OpenStreetMap through the Leaflet library, enabling dynamic visualization of available rental properties on an interactive map. This feature helps users explore properties based on geography, improving decision-making and navigation.

By offering a streamlined, transparent, and secure rental process, the system benefits both tenants and landlords, making house renting more accessible and efficient.

Keywords**:** Rental System, House Management, AI Recommendation, OpenStreetMap, Accessibility, Django.

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ABBREVIATION

AI :Artificial Intelligence

API :Application Programming Interface

B2C :Business-to-Consumer

CRUD :Create, Read, Update, Delete

CSS :Cascading Style Sheets

DFD :Data Flow Diagram

DSF :Django Software Foundation

DMS :Database Management System

HTML :Hypertext Markup Language

HTTP :Hypertext Transfer Protocol

JS :JavaScript

JSON :JavaScript Object Notation

MTV :Model Template View

OTP :One Time Password

PDF :Portable Document Format

SQL :Structured Query Language

UI :User Interface

UX :User Experience

CHAPTER 1: INTRODUCTION

1.1 Background

The rental housing market has seen significant changes over the past few years, primarily due to advancements in technology and evolving consumer expectations. Digital platforms are increasingly becoming the preferred medium for landlords and tenants to manage rental transactions, as they offer greater convenience, efficiency, and transparency. In contrast, traditional methods, such as manually listing properties and conducting in-person viewings, are often time-consuming and lack clarity [1]. This shift in user needs has created a growing demand for modern online platforms that simplify these processes while improving user satisfaction.

However, despite the progress made, several challenges still persist. A key issue in many rental platforms is the lack of secure payment systems, which raises concerns about the safety of transactions and undermines user trust [2]. While our current system does not yet implement secure payment integration, we acknowledge its importance as a future enhancement. Additionally, the absence of AI-powered recommendation systems in most platforms limits users' ability to efficiently find properties that meet their specific preferences, often resulting in lengthy and frustrating property searches [3]. Our system addresses this gap by incorporating personalized property suggestions. Furthermore, many platforms fail to provide effective search and filtering options, making it difficult for users to navigate through extensive listings and discover properties that match their criteria.

To address these issues, the proposed House Rental System seeks to create an efficient, secure, and user-friendly platform. The system includes features such as AI-driven recommendations to enhance user experience by offering personalized property suggestions tailored to individual preferences. Additionally, advanced search and filtering tools help users quickly and easily find properties that match their needs. By ensuring transparency and simplifying the overall process, the system aims to build trust and improve convenience for both tenants and landlords.

Security and trust are critical components of the rental market, as fraud and scams remain significant concerns. Many platforms currently lack proper verification processes, leaving users vulnerable [4]. To combat this, the proposed system includes OTP verification during user registration and authentication, helping ensure that only genuine users access the platform. Additionally, the system incorporates transparent communication tools between tenants and landlords, further creating a safe and reliable environment for all users.

To enhance location awareness and visual context, the system integrates OpenStreetMap using the Leaflet.js library. This allows users to view rental properties directly on an interactive map based on geographic data. By visualizing properties spatially, users can make more informed decisions based on proximity to landmarks, neighborhoods, and amenities—adding a powerful geospatial dimension to the house-hunting experience.

In conclusion, the House Rental System aims to transform the way landlords and tenants interact by addressing key issues in the rental market. Through the integration of advanced technology, a focus on user-centric design, and a commitment to security, this platform has the potential to create a seamless, efficient, and trustworthy rental experience for all users.

1.2 Problem Statement

The proposed system aims to address the challenges faced by both landlords and tenants in the house rental market by developing an online HOUSE RENTAL SYSTEM that not only streamlines the rental process but also enhances user experience through advanced technological features. Currently, many existing platforms lack seamless interaction through responsive and modern web technologies, resulting in usability issues across devices. Additionally, the absence of AI-powered recommendation systems limits users' ability to find properties that best match their preferences and needs, resulting in a time-consuming and often frustrating search process. Furthermore, inefficient search and filtering options hinder users from easily discovering suitable properties, making it difficult for them to navigate the vast array of available listings.

1.3 Objectives

The objectives of the proposed system are as follows:

i. To build a responsive house rental platform that enables seamless interaction through modern web technologies.
ii. To integrate AI-powered recommendation systems into the platform.
iii. To enable efficient search and filtering options to simplify the discovery of properties.

1.4 Scope and Limitations

The project scope defines the description of the work that is required in delivering the Rental House Management System. The following are the scopes of work during the course of the project:

- User-Friendly Platform: Develop an intuitive UI and User Experience UX-oriented online platform for both landlords and tenants to navigate easily.

- Property Listings: Allow landlords to list available rental properties with detailed descriptions, images, and pricing using CRUD operations.
- Search and Filtering: Implement advanced search and filtering options using SQL queries to help tenants find properties that meet their specific criteria.
- AI-Powered Recommendations: Integrate AI algorithms to provide personalized property recommendations based on user preferences and behaviors.

The project is primarily focused on developing AI-powered recommendations to enhance property discovery for users, which may limit the initial incorporation of other advanced features like advanced analytics for market trends, comprehensive user profiles, and integrated secure payment options.

1.5 Applications

This web-based application streamlines renting by connecting tenants and landlords, offering convenience, transparency, and efficiency through an online platform.

Core Features

1. **Online Property Listings & Search**

   - Tenants browse properties by location, budget, size, and furnishing.
   - Landlords post listings with details, images, rent, and availability.

2. **AI-Based Recommendations**

   - Personalized suggestions based on user preferences and behavior.

3. **Booking Management**

   - Tenants send booking requests and track their status (pending, accepted, rejected).
   - Landlords review and approve or reject requests to manage rentals.

4. **Authentication and Security**

   - Users sign up and log in to access features.
   - OTP verification and role-based access ensure secure use.

5. **Interactive Map Integration (OpenStreetMap)**

   • Properties are displayed on an interactive map using **Leaflet.js** and **OpenStreetMap**.

CHAPTER 2: LITERATURE REVIEW

2.1 Overview of Existing Systems

Recent research in house rental systems demonstrates a clear evolution towards technology-driven solutions that prioritize user experience, security, and efficiency. These developments address various challenges in traditional rental processes while introducing innovative approaches to property management and tenant-landlord interactions.

2.1.1 Local Platforms

a. Ropani.com:

Ropani.com is a property-focused platform in Nepal that offers listings for buying, selling, and renting real estate. It is widely used for property-related transactions in the country. Items Covered: Real estate (houses, apartments, land for sale or rent)[5].

Limitations:

- Narrow Focus: Exclusively deals with real estate, leaving out other categories like vehicle or equipment rentals.
- Market Competition: Faces strong competition from other real estate websites in Nepal.

b. Gharbeti:

Gharbeti is a platform primarily dedicated to property rentals, including apartments, houses, and rooms. It focuses on helping tenants and landlords connect.

Items Covered: Residential properties (apartments, houses, rooms), Commercial properties (offices, shops)[6].

Limitations:

- Limited Scope: Only covers property rentals, without offering other types of rentals like vehicles or appliances.
- Target Audience: Mainly focused on people looking for housing, which limits its appeal to a broader market.

c. Rental Nepal:

Rental Nepal is a platform that provides various rental services, including property, vehicles, and equipment. It connects users to rental listings for both personal and business needs.

Items Covered: Property rentals (apartments, houses, offices), Vehicle rentals (cars, bikes), Equipment rentals (tools, machinery) [7].

Limitations:

- Market Fragmentation: Offers a wide range of categories, which can make the user experience feel disorganized.
- Limited Availability: Availability of certain rental items, especially vehicles and equipment, can be scarce depending on location.

d. Erentalservice:

Erental service is an online platform that specializes in providing rental services for various types of properties and equipment. It enables both tenants and landlords to manage their rentals efficiently [8].

Items Covered: Property rentals (residential and commercial), Equipment rentals (tools, machinery), Vehicle rentals

Limitations:

- Limited Categories: Primarily focused on property and equipment, without options for other rentals such as fashion or event-related items.
- User Experience: The platform might be difficult to navigate for first-time users due to its complex features.

Table 2. 1 Comparative Analysis of Existing Platforms

Platform	Technologies Used	Features	Strengths	Drawbacks
Ropani.com	PHP, Google Maps API, Basic CMS	Comprehensive property listings, map integration, property valuation tools, real estate news updates	- Easy-to-use interface - Caters to both buyers and sellers - Provides real estate insights	- Limited advanced search options -No AI-powered recommendations - Lacks automated communication
Gharbeti	PHP, Google Maps API,	Tenant landlord	Simple,intuitive platform - Free	- Limited automation for

		matching,rental agreements, basicproperty management tools	for basic use - Focused on rental properties	tenant-landlord interactions
Rental Nepal	PHP, Laravel Framework, Database Systems	Verified rental listings, property management tools, tenant screening services	- Trusted verified listings - Supports tenant screening - Reliable for rental housing	- Limited scalability for large property markets - Few features for property buyers or investors
Erentalservice	ASP.NET, Cloud Hosting, SQL Server Database	Online rental agreements, tenant and landlord profiles, payment tracking	- Secure payment tracking - Supports rental agreements digitally - User profiles for better tracking	- Limited AIbased recommendati ons - Lacks advanced features like virtual tours or property suggestions

2.1.2 Case Study: AI-Based Algorithms

a. Content Based Filtering Using Cosine Similarity

Content-based filtering, specifically using cosine similarity, is an excellent technique for recommending rental properties like houses, rooms, or apartments, as it focuses on comparing properties based on their specific attributes. In this method, properties are represented as vectors, where each dimension corresponds to a feature such as price, location, number of rooms, square footage, and amenities. Cosine similarity is then used to measure the similarity between the user's preferences and the available properties in the database. The formula for cosine similarity is given by:

Cosine Similarity = (A · B) / (||A|| ||B||)

- A · B represents the dot product of vectors A and B,
- ||A|| and ||B|| represent the magnitudes (or Euclidean norms) of vectors A and B

```
Cosine Similarity Matrix:
[0.   0.25 0.   ... 0.   0.   0.  ]
Recommended Properties: ['Elegant 2 Bed Apartment in Submetropolitan city-4, Sunsari', 'Modern
2 Bed House in Mahendra Nagar, Sunsari', 'Cozy 2 Bed House in Dharan-12, Sunsari', 'Charming 2
Bed House in Itahari, Sunsari', 'Spacious 3 Bed House in Dharan, Sunsari']
[06/Jul/2025 18:37:45] "GET /property/favorites/ HTTP/1.1" 200 9663
```

By calculating the cosine of the angle between two vectors, the system quantifies how closely a property's features match the user's preferences. A value closer to 1 means the property closely aligns with the user's preferences, and 0 indicates no similarity. This method ensures that the recommended rental properties are highly relevant to the user, making it a powerful tool for rental platforms. As highlighted in the paper [9], cosine similarity ensures that recommendations are tailored to user preferences, offering a personalized experience in finding rental properties.

Pros :

Overcomes Cold-Start Problem: Content-based methods are especially useful for new users, as they do not require interaction history. New users can specify their preferences (such as budget, preferred location, or room size), and the system can generate recommendations based purely on these preferences. This approach is particularly beneficial in the context of rental properties, where the available listings change frequently. The ability to generate immediate recommendations, even without historical user data, helps rental platforms quickly engage new

users. This method avoids the cold-start problem, making it ideal for rental platforms with frequent new users.[10]

Personalized Recommendations: Content-based filtering provides highly tailored recommendations by matching user preferences with property features. For example, if a user specifies a preference for an apartment within a certain price range or in a specific neighborhood, the system will prioritize properties that meet these criteria. This level of personalization is especially important in the rental market, where users are looking for specific characteristics such as the number of rooms, type of property (house, apartment, etc.), and amenities (e.g., parking, gym, etc.).

Scalability and Adaptability: Content-based filtering scales well as new rental listings are added to the platform. When a new property is listed, the system can update the similarity scores based on the property's features and adjust recommendations accordingly. This makes the system adaptable to the ever-changing rental market, where new listings are constantly being added. The ability to incorporate new properties into the system quickly ensures that the recommendations stay fresh and relevant.

Simple to Explain and Transparent: One of the advantages of content-based filtering is its transparency. Users can easily understand why certain properties are recommended to them, as the recommendations are based on their specific preferences and the features of the properties. This builds trust with the users, as they can see the rationale behind the system's suggestions.

Cons :

Limited by Available Features: The effectiveness of content-based filtering depends heavily on the features available to describe the properties. If important user preferences (like proximity to public transport or aesthetic qualities) are not captured, the recommendations may not be as useful.

Overfitting: The system may over-recommend properties that are too similar to the user's stated preferences, leading to a lack of diversity in suggestions.

Feature Engineering is Critical: The quality of recommendations is dependent on the choice of features. Poorly chosen or incomplete features may lead to inaccurate or suboptimal recommendations.

Lack of Exploration: Since the system focuses on matching user preferences, it may limit exploration by only suggesting properties that closely match the user's preferences, possibly missing out on other potentially interesting option.

2.2 Significance of Proposed Work

The identified gaps in existing systems, particularly in the Nepalese context, present opportunities for significant improvements in rental platform development. By addressing these limitations through enhanced verification systems, and improved user interactions, and advanced property filtering mechanisms, new solutions can better serve the specific needs of the local market. Additionally, incorporating proven technological advances from global research, such as AI-powered recommendations can position new platforms as innovative leaders in the rental market.

CHAPTER 3: METHODOLOGY

The project follows a systematic approach to design and implementation, ensuring the development of a robust and user-friendly HOUSE RENTAL SYSTEM focused on enhancing the rental experience for both landlords and tenants. The methodology incorporates requirements gathering, a case study analysis of existing research, and the design of an AI based recommendation algorithm to improve property matching. Additionally, modern technologies such as Django for back-end development and HTML, CSS and JS for front-end design will be utilized to create an efficient and responsive platform. This structured approach aims to address the challenges of user satisfaction and system effectiveness while ultimately providing a seamless rental management experience.

There are 4 different stages which are flow of project, system design, software and hardware requirement and testing and maintenance.

3.1 Flow of Project

The flow of a **House Rental System Project** involves several key stages:

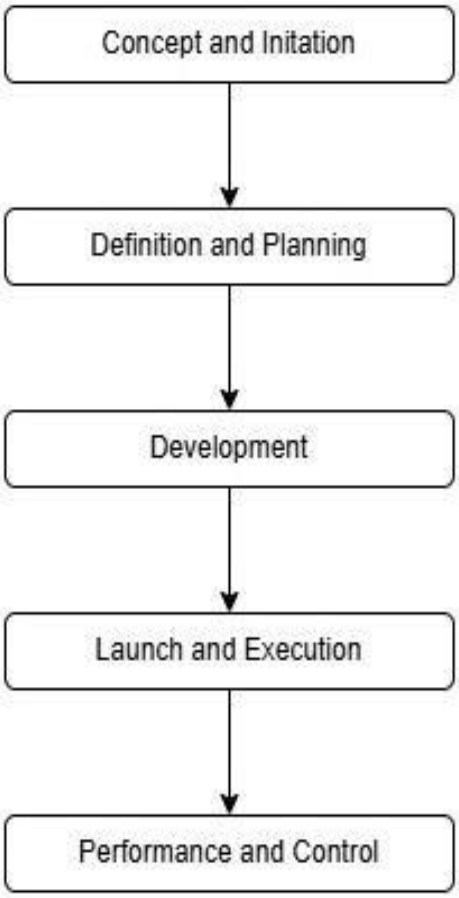


Figure 3. 1 Flow of project

3.1.1. Concept and Initiation

The primary goal of the HOUSE RENTAL SYSTEM is to provide a platform where property owners (landlords) can list their properties, and potential renters (tenants) can search and book rentals seamlessly. Stakeholders include:

Landlords: Property owners listing their rentals.

Tenants: Individuals searching for rental properties.

Objectives: Manage user accounts and property listings. Monitor platform performance to enhance user experience.

Scope: The project encompasses the development of a web application with features for property management, user interaction, and intelligent property discovery.

3.1.2. Definition and Planning

In our HOUSE RENTAL SYSTEM project, we aim to create a software solution that streamlines and automates the rental process for residential properties. This system will help property owners, managers, and tenants manage their rental operations efficiently.

Key Steps:

Requirements Gathering: Identify the needs of landlords, tenants, and administrators.

Project Scope: Define features like property listings, booking management.

Timeline: Set a project timeline with milestones to track progress.

Resource Allocation: Determine the necessary resources, including team members and budget.

3.1.3. Development

In our "HOUSE RENTAL SYSTEM" project, we will utilize the following technologies:

1. **HTML:** Serves as the foundation for the user interface, structuring and displaying content such as property listings, search features, and navigation menus.
2. **CSS:** Controls the visual appearance and layout of the web pages, ensuring the UI is attractive, responsive, and user-friendly. It enhances property cards with styles like shadows, borders, and hover effects.

3. **JavaScript**: Enables dynamic functionality and interactive features, allowing users to search and filter properties based on criteria like location and price. It also handles form validation and the booking process.
4. **Django:** A Python-based web framework that follows the model- views- template (MVT) architectural pattern. It will be used to interact with the MySQL database for CRUD operations and to implement a recommendation system for personalized property suggestions.
5. **Recommendation System:**

   For the AI-based recommendation system, we will utilize **Pandas** and **scikit-learn** to develop a model that suggests houses based on user preferences. **Pandas** will be used for data preprocessing, organizing, and analyzing user and house data.

3.1.4. Launch and Execution

A well-structured plan ensures the smooth launch and operation of a HOUSE RENTAL SYSTEM. It ensures all the features like property management, tenant management. Once the development of our HOUSE RENTAL SYSTEM, we can verify that the interface is userfriendly, responsive, and meets accessibility standards. We will ensure compatibility, conduct testing, and gather valuable user feedback to make necessary improvements.

3.1.5. Performance and Control

The performance and control phase of the house rental system project ensures that the system operates efficiently, meets performance benchmarks, and adheres to the defined quality standards. We will collect feedback from users (landlords, tenants) through surveys, reviews, or ratings.

3.2 System design

Once the requirements are collected and analyzed, they must be given a proper structure. In this phase the architecture of the project will be designed based on the requirements collected in previous phase. In this phase many architecture diagrams such as ER diagram, DFD (data flow diagram), Use Case, system flow diagram are used .

3.2.1 System Flow:

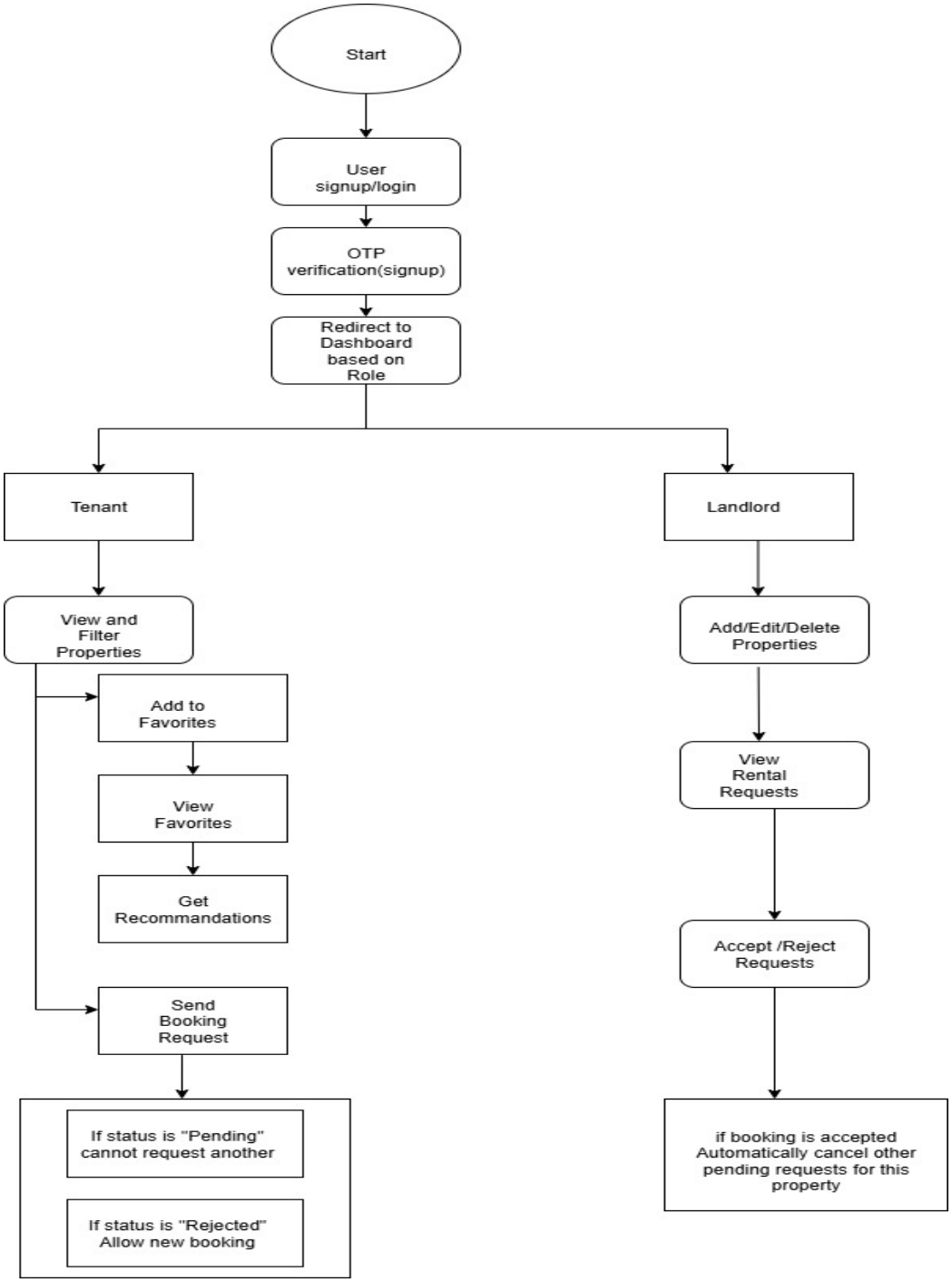


Figure 3. 2 System Flow diagram

3.2.2 E-R diagram

E-R Diagram stands for Entity Relationship Diagram, also known as ERD is a diagram that displays the relationship of entity sets stored in a database. In other words, ER diagrams help to explain the logical structure of databases. ER diagrams are created based on three basic concepts: entities, attributes and relationships. ER Diagrams contain different symbols that use-rectangles to represent entities, ovals to define attributes and diamond shapes to represent relationships.

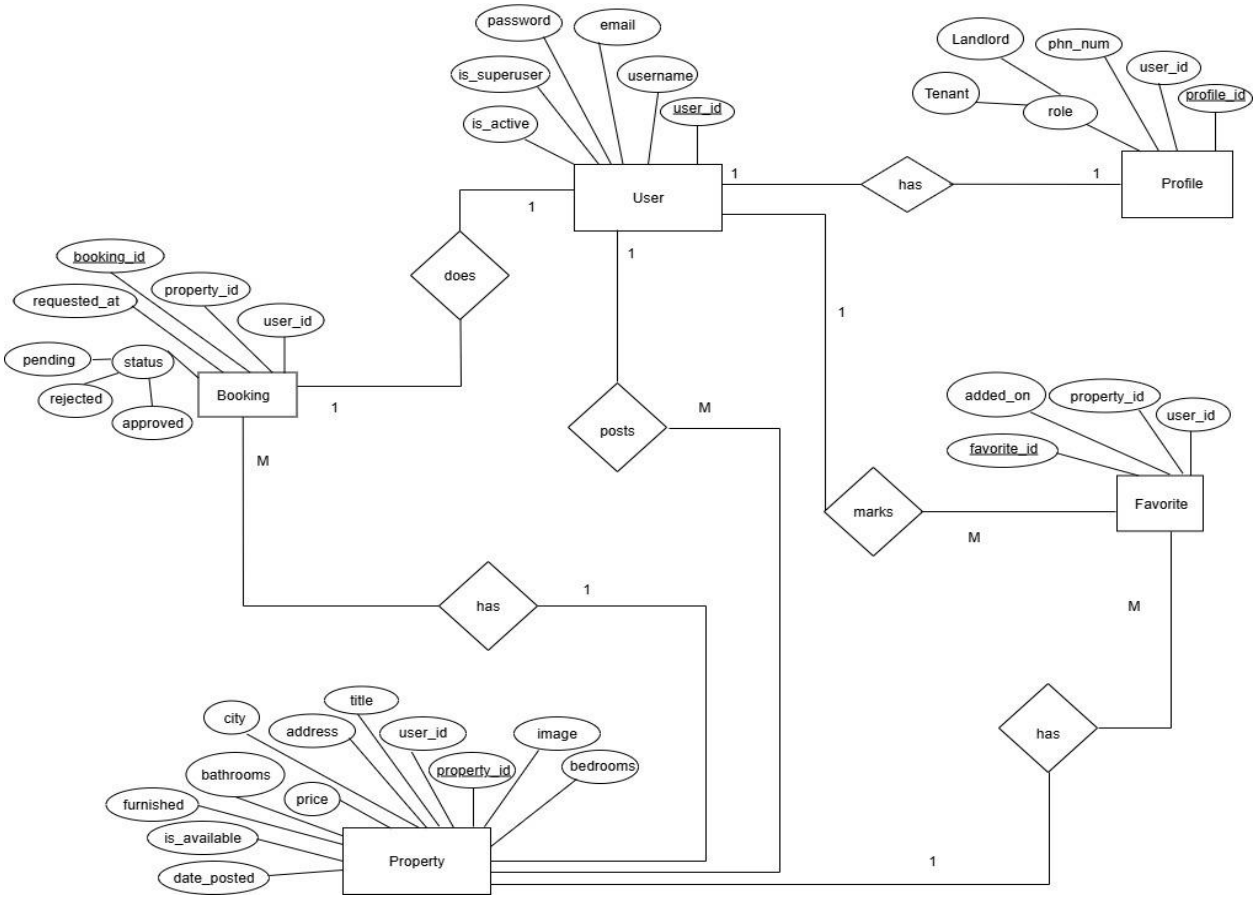


Figure 3. 3 ER diagram

3.2.3 Data Flow Diagram (DFD)

A Data Flow Diagrams (DFDs) are important tools for our House Rental System project. DFDs visually show how data moves in our system, helping us find and fix issues to make it work better . They give us a clear picture of how our system is built and what it does, making it easier for our team to work together. DFDs can be used at different levels, from big pictures to small details, to help us design and improve our system.

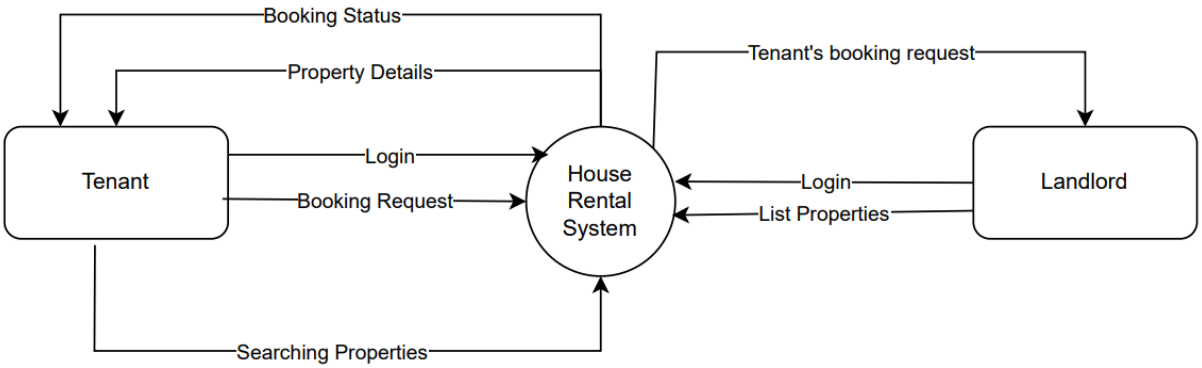


Figure 3. 4 Level 0 DFD

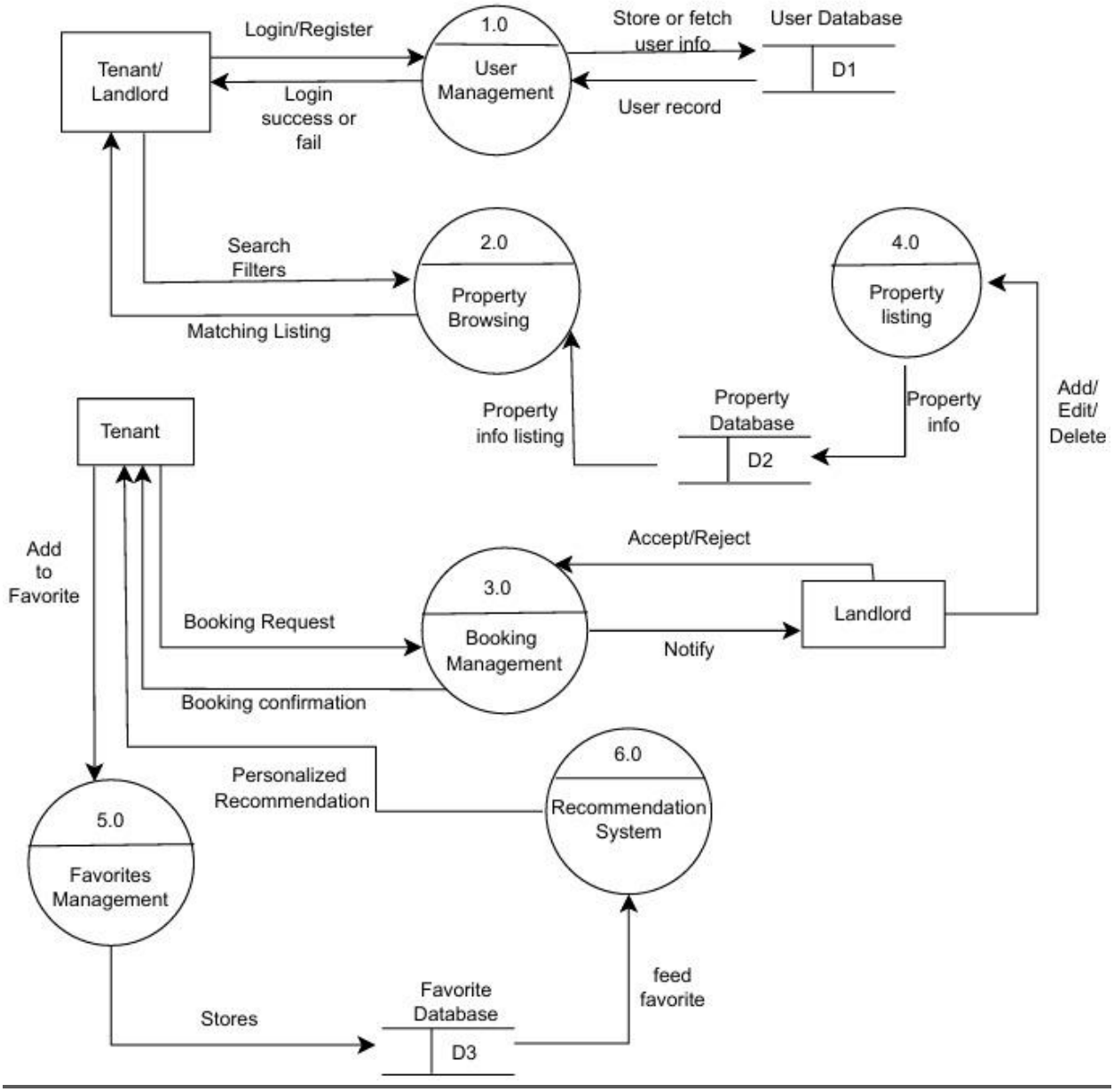


Figure 3. 5 Level 1 DFD

3.2.4 Use case Diagram

A use case diagram is a graphical depiction of a user's possible interactions with a system. A use case diagram shows various use cases and different types of users the system has and will often be accompanied by other types of diagrams as well.

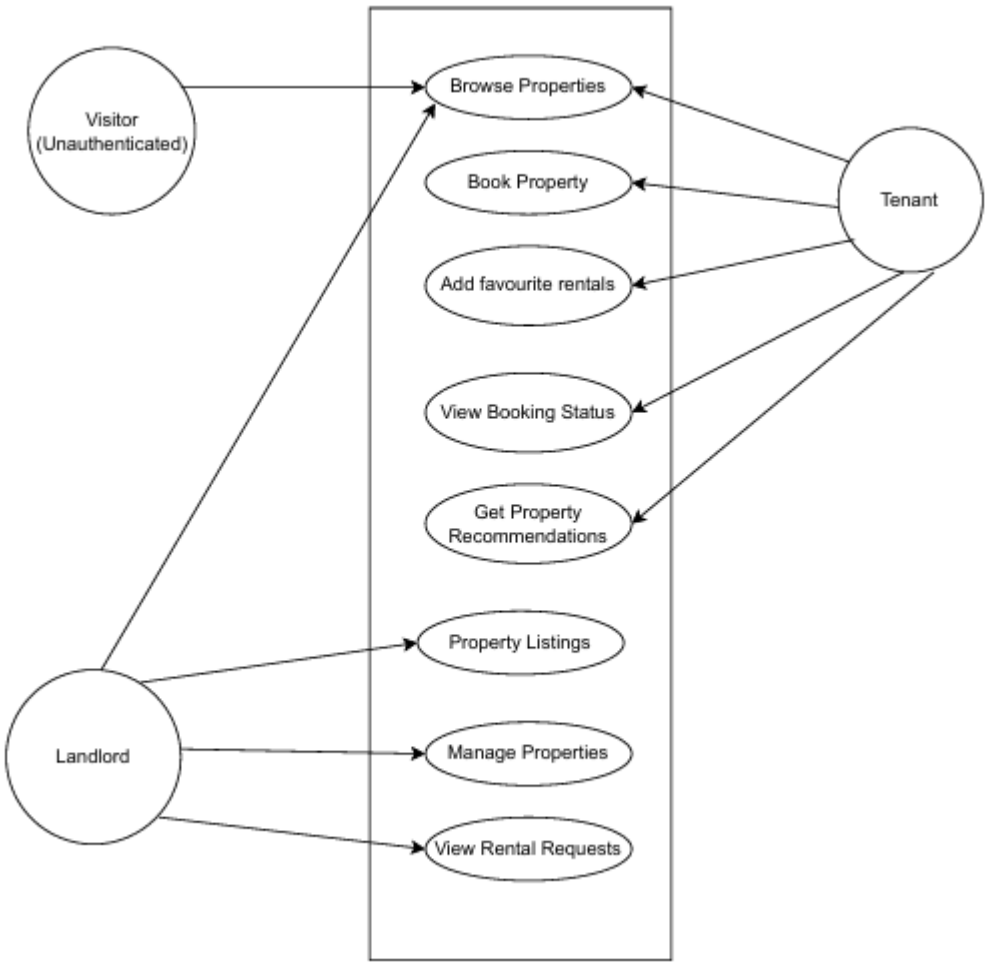


Figure 3. 6 Use Case Diagram

3.2.5 Activity Diagram:

An activity diagram for a house rental system illustrates the flow of activities from both the tenant and landlord perspectives, detailing the key interactions with the system. It visualizes how users interact with features such as property browsing, booking, and approvals.

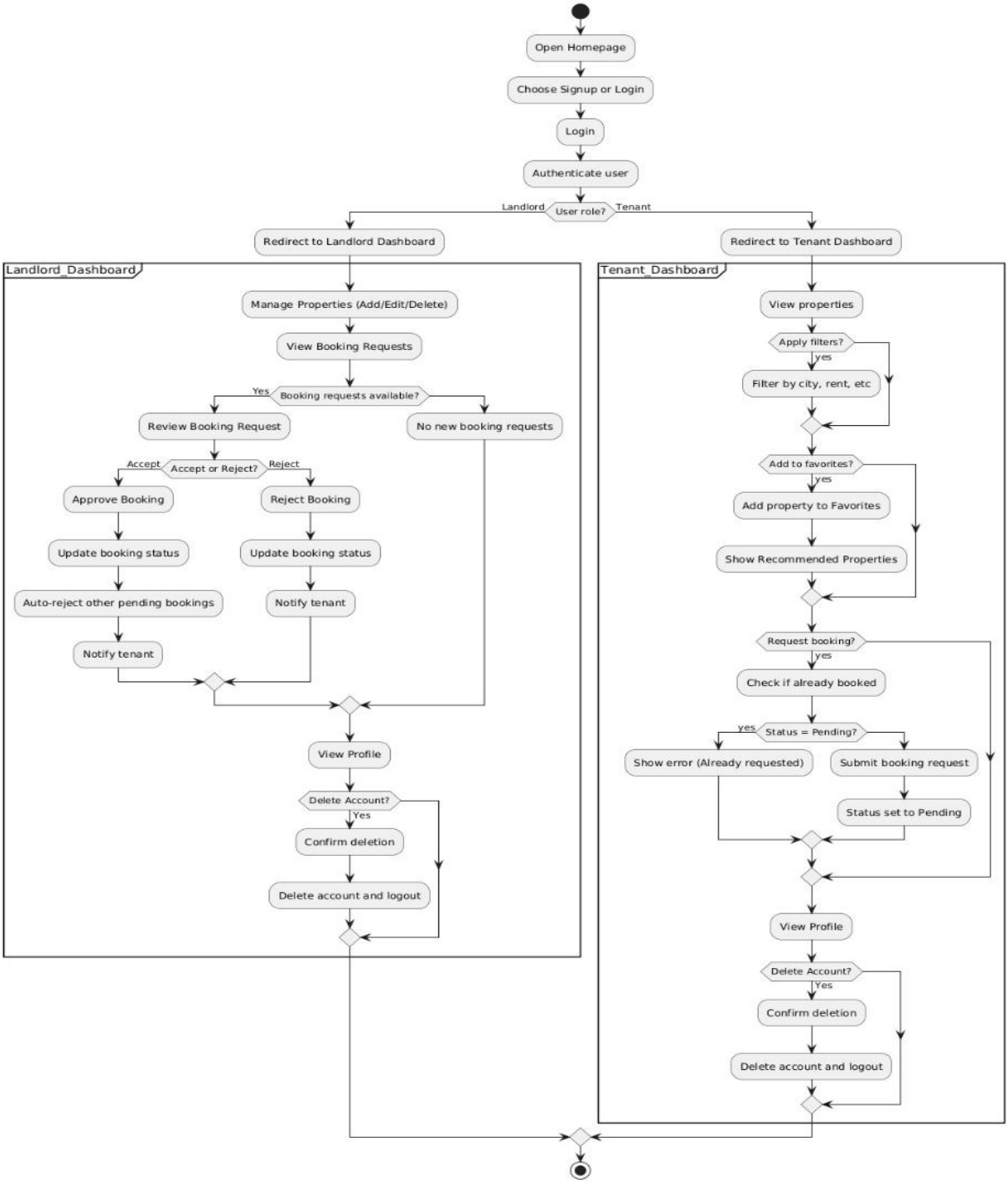


Figure 3. 7 Activity Diagram

3.2.6 Sequence Diagram:

The sequence diagram shows how a tenant searches, views listings, and sends a booking request. The system notifies the landlord, who accepts or rejects it. The decision is then communicated back to the tenant, illustrating the key interactions and booking updates in real time.

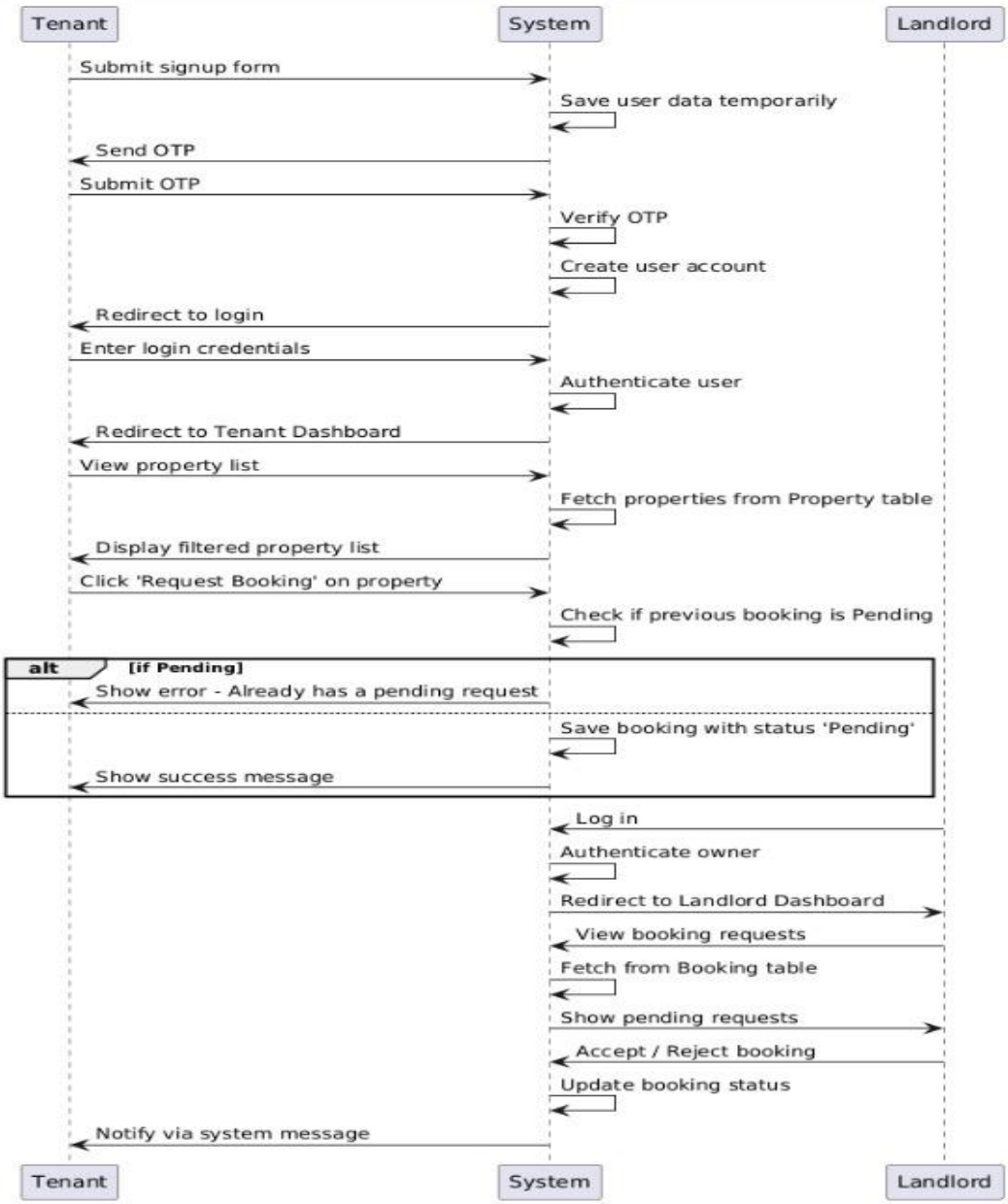


Figure 3. 8 Sequence Diagram

3.3 Software and hardware requirement:

For our HOUSE RENTAL SYSTEM project, the hardware and software requirements are as follows:

Hardware Requirements:

1. **Mobile Devices:** Any smartphone or tablet with an internet connection and a modern web browser.
2. **Computers:** Any desktop or laptop computer with an internet connection and a modern web browser. The system should be compatible with both Windows and mac OS operating systems.
   Software Requirements:
3. **Operating System:** Compatible with various operating systems, including Windows, mac OS, Android, and iOS.
4. **Web Browsers:** Support for popular web browsers such as Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari.

By ensuring compatibility with a wide range of devices and operating systems, our HOUSE RENTAL SYSTEM can be accessed and used seamlessly by users on the smartphones, computers, and other devices.

3.4 Testing and maintenance

Although automated unit tests were not fully implemented in `tests.py`, the system was manually tested throughout development. Functionality such as user signup, login, OTP verification, profile deletion, property listing, and booking flow were debugged and verified using realistic inputs. The table below summarizes the core test cases considered during development.

In terms of maintenance, future improvements include regular monitoring for bugs, updating UI components, adding secure payment integrations, and enhancing the AI recommendation module for better user experience.

Test Case ID	Feature	Input	Expected Output	Result
TC-01	User Signup	Valid username ,email, password	Redirected to OTP verification page	Pass
TC-02	OTP verification	Correct OTP entered	Account verified and redirected to login page	Pass
TC-03	Login	Verified user enters correct email and password	User redirected to dashboard based on role (tenant/landlord)	Pass
TC-04	Property Booking	Tenant selects property and submits booking request	Booking created with status “Pending”	Pass
TC-05	Booking Cancellation	Tenant cancels a pending booking	Removed from active bookings	Pass
TC-06	Rebooking After Rejection	Tenant rebooks the property after rejection	Booking allowed and new request sent	Pass
TC-07	View Landlord Info	Booking status is “Accepted”	Landlord’s name, phone, and email displayed	Pass
TC-08	Duplicate Booking Attempt	Tenant with active (Pending/Accepted) booking tries again	Error message shown, booking not allowed	Pass

CHAPTER 4: EPILOGUE

4.1 Work Completed

We are pleased to highlight the noteworthy accomplishments achieved during the development and implementation of the Rental App system:

A. Frontend Design

Front-end design helps us create component to seamless interact with users. Our project serves as the visual and interactive interface that connects users the core functionalities of the platform. We have used HTML, CSS, JavaScript here for responsive design, navigation ease and user-friendliness.

Key user interface elements include property search, filters, user login, and property listings views that are built using Django's template engine, promoting dynamic web page.

Key Features:

- **Search and Filtering:** Users can search for property of their type, location and price through a clean, intuitive UI.
- **User Authentication Interface:** Registration and login pages ensure secure access to user-specific feature.
- **Responsive Layout:** Seamlessly adapt to both desktop and mobile device**.**
- **Property Listings Display:** Landlord can showcase their properties with relevant details including images, descriptions, and pricings.

B. Backend Design

Our project back-end architecture is developed by using Django framework, adhering MVT (Model-View-Template) design pattern. It helps to organize the codebase and workflow of a web application handling logic, data processing, and communication between the user interface and database.

Technologies used:

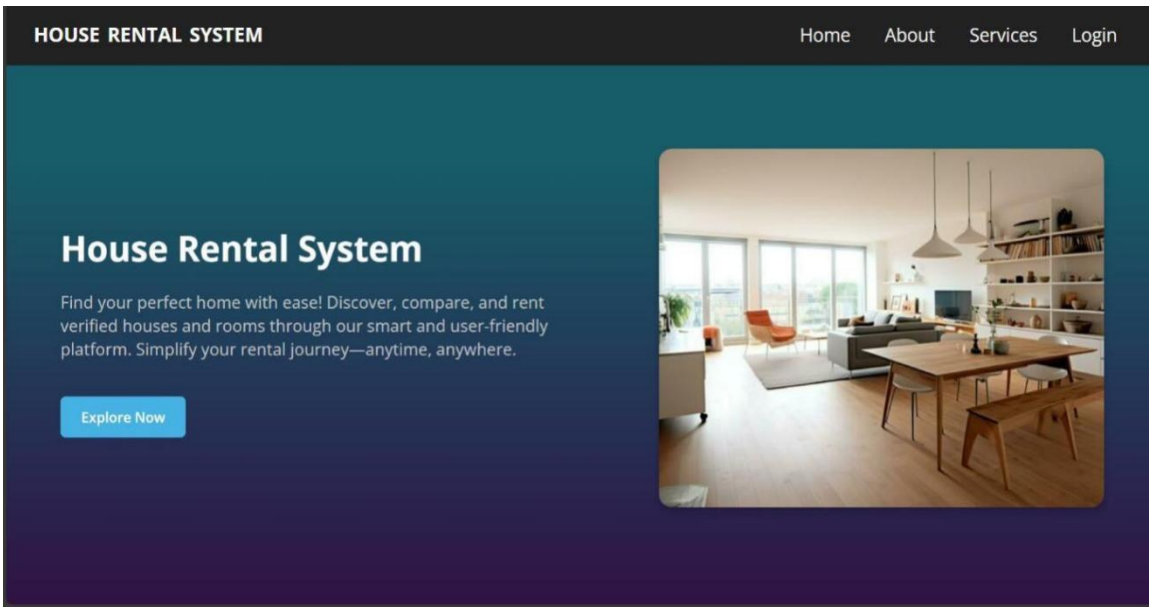
Django(Python): Django Framework was used, including Django ORM for database interactions, the Messages Framework for user feedback, and built-in authentication mechanisms with OTP integration for secure login.

MYSQL: Stores user data, property listings, bookings, and favorites.

We have attached some of our designs below:

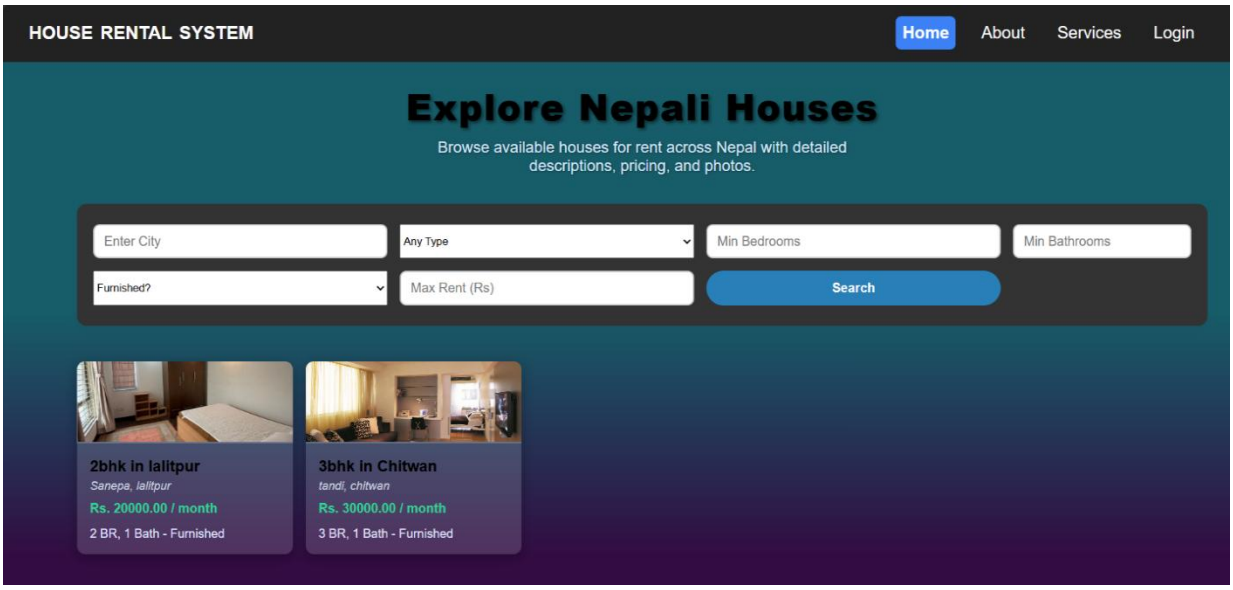
1. Landing Page :

The landing page of the House Rental System features a responsive layout with navigation links and a clear introduction to the platform. It includes a call-to-action button and property image, providing users an easy and engaging starting point.



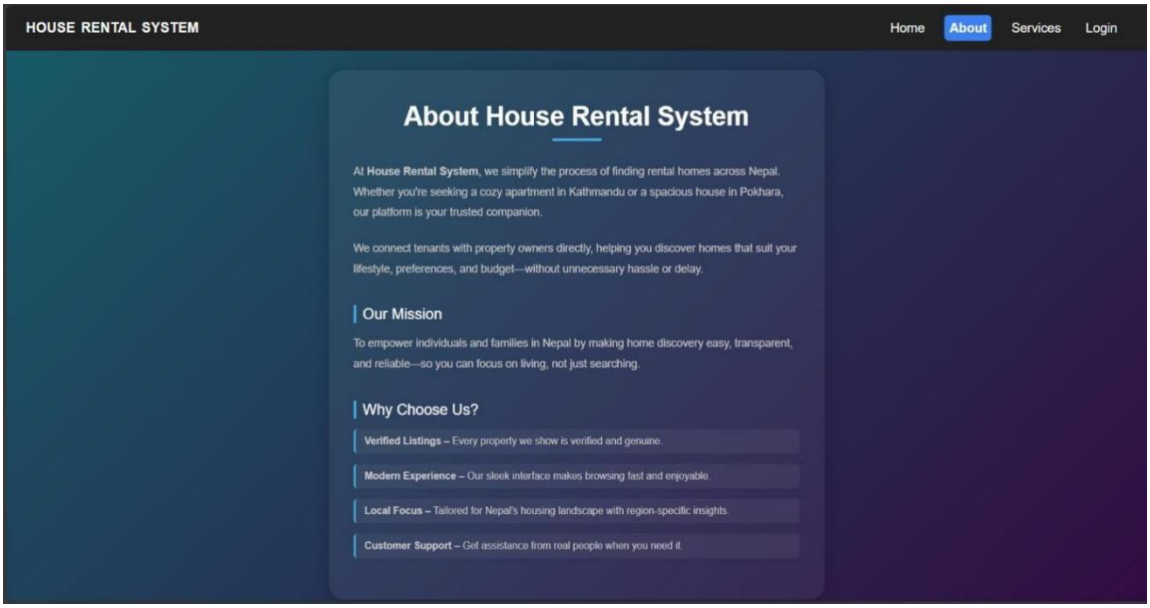
2. Home Page:

The figure shows the House Rental System's home page with a search bar for filtering properties by city, type, bedrooms, bathrooms, furnishing, and rent. Below, matching listings display key details and images. The top navigation bar offers quick access to Home, About, Services, and Login, making property browsing simple and efficient.



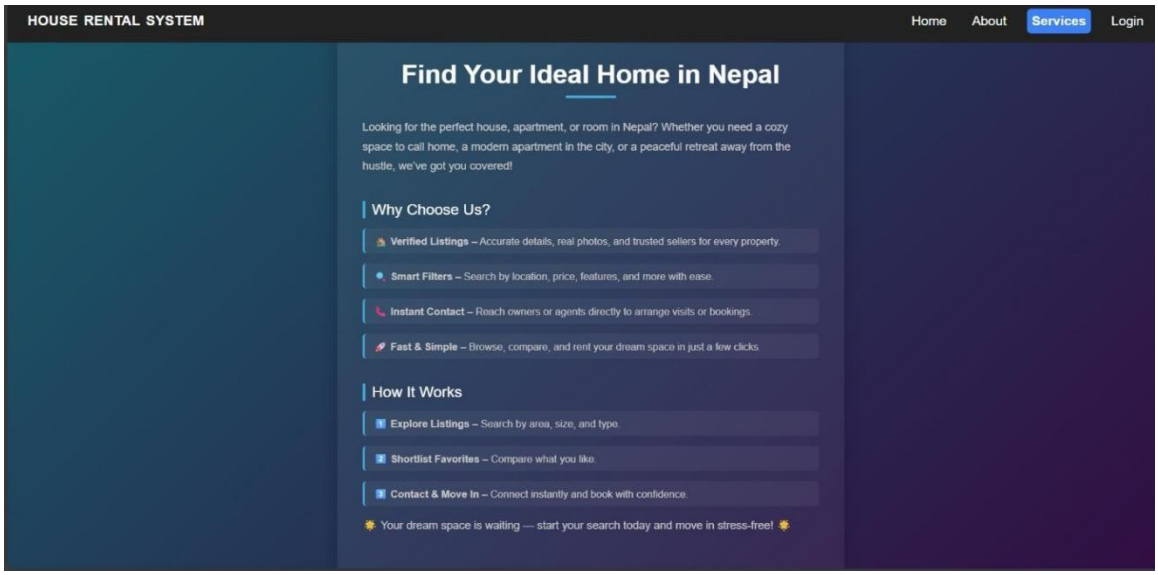
3. About Us page :

The figure displays the "About Us" page of the House Rental System, highlighting its mission to connect tenants with verified properties in Nepal. It emphasizes transparency, accessibility, and features like verified listings, user-friendly design, local focus, and real-time support, all presented in a clear, trustworthy layout.



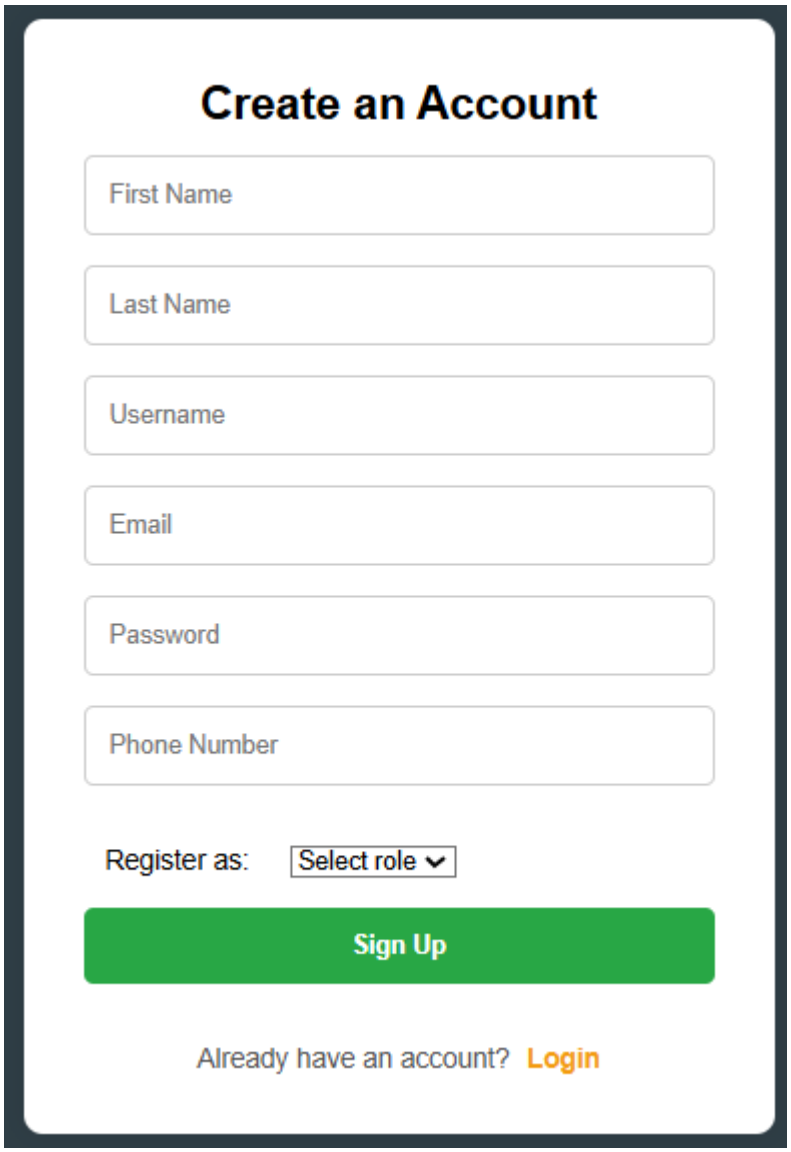
4. Service page:

The figure shows the Services page of the House Rental System, presenting key features like verified listings, smart filters, instant contact, and easy booking. The “How It Works” section guides users through searching, shortlisting, and booking properties. The design uses clear sections and icons to highlight convenience and streamline the rental process.



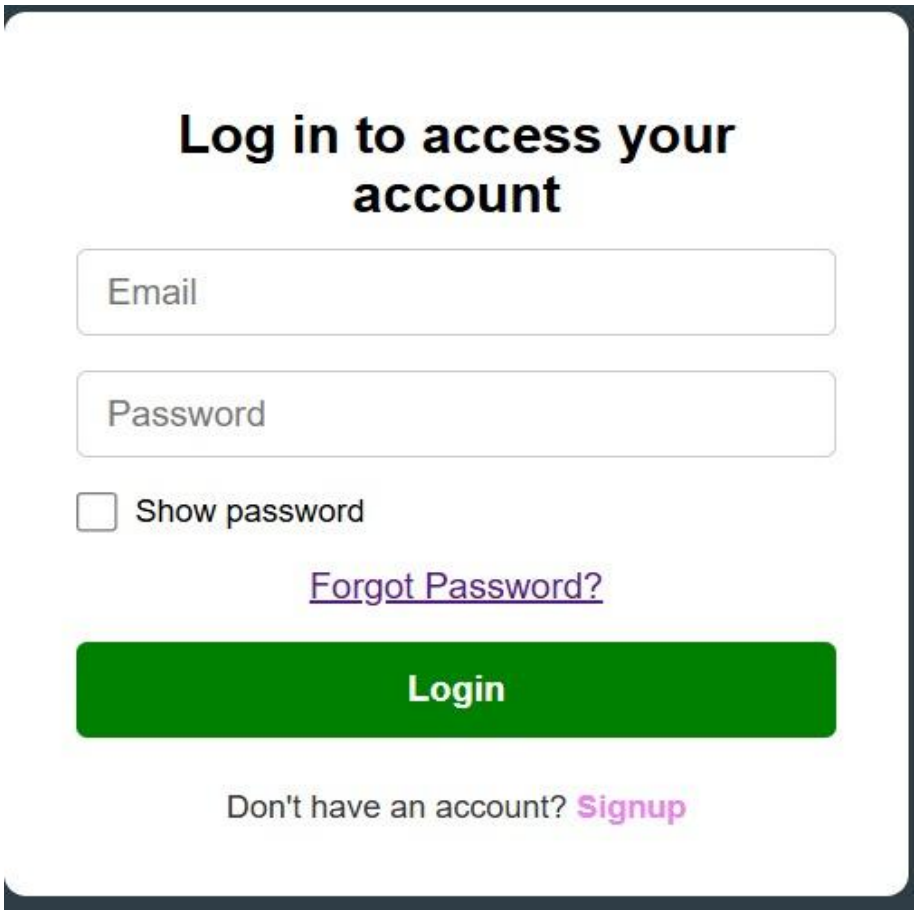
5. Sign-up page :

The figure shows a user registration form with fields for name, username, email, password, phone, and role selection. A green “Sign Up” button submits the form, while a link below directs existing users to the login page. The design is clean and user-friendly.



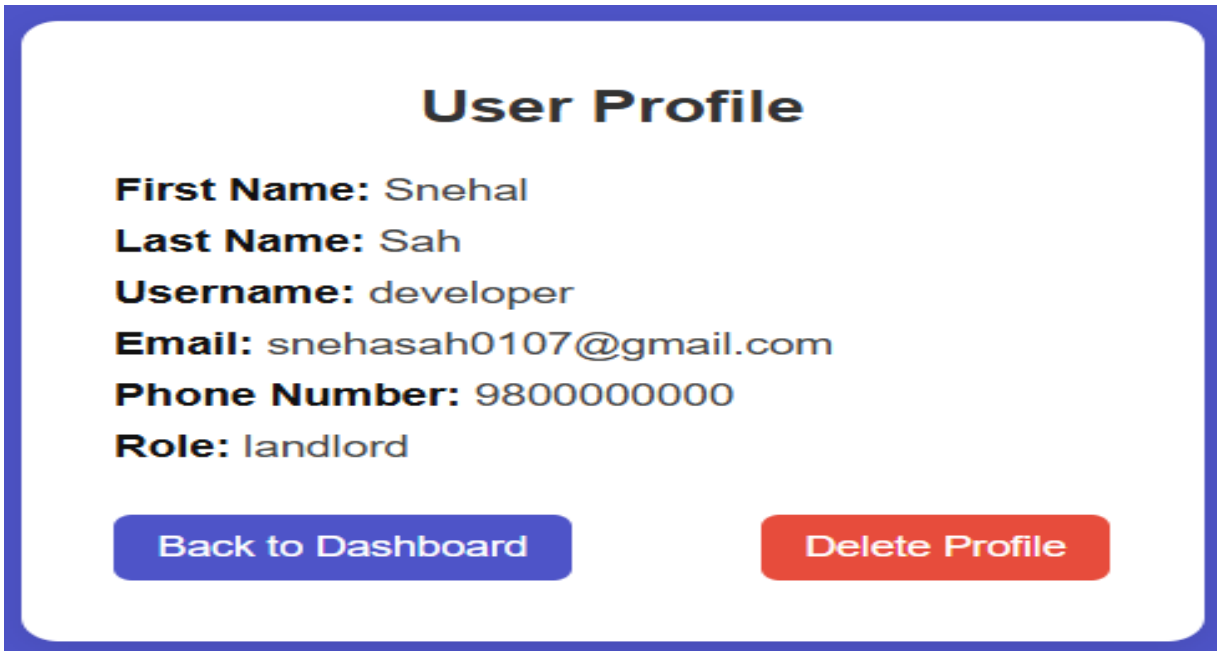
6. Login page :

The figure shows a web login form with email and password fields, a password visibility toggle, and a green "Login" button. Links for password recovery and registration are provided below. The design focuses on clarity, usability, and security.



7. Profile page:

The figure shows a user profile page displaying personal details like name, username, email, phone, and role. It includes buttons to return to the dashboard and to delete the profile, focusing on simple account management.



8. Property List Database :

The figure presents a tabular view of property listings from a backend database, likely part of a property rental management system. Each row corresponds to a rental unit and includes attributes such as property title, type, address, city, number of bedrooms and bathrooms, furnishing status, monthly rent, availability date, and associated image file path. Additional fields include availability status (is_available) and a foreign key (landlord_id) indicating the listing owner. This structure supports dynamic rendering of rental data within a web application.

id	title	property_type	address	city	bedrooms	bathrooms	furnished	monthly_rent	available_from	image	is_available	landlord_id
1	2bhk in lalitpur	Apartment	Sanepa	lalitpur	2	1	1	20000.00	2025-07-03	property_images/Screenshot_2025-06-22_170107_2SxBP...	1	5
2	3bhk in Chitwan	Apartment	tandi	chitwan	3	1	1	30000.00	2025-07-03	property_images/room_v3FzUqR.jpg	1	5

9. Booking DataBase:

The image displays a database table entry representing a maintenance or service request. The table consists of five fields: id, status, requested_at, property_id, and tenant_id. This table structure is indicative of a relational database used in property management systems to track and manage tenant service requests.

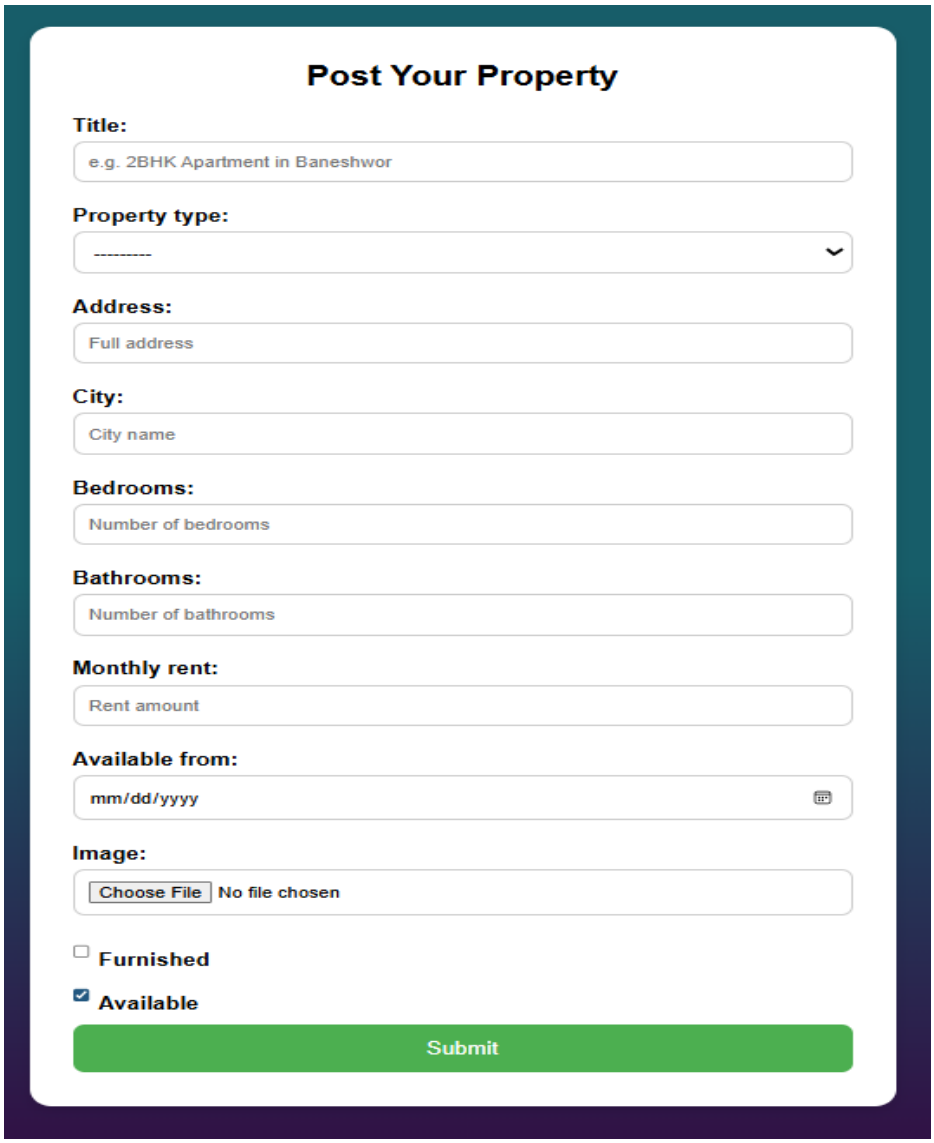
			id	status	requested_at	property_id	tenant_id	note
Edit	Copy	Delete	12	Accepted	2025-07-05 16:40:23.102400	1	7	hey i am so so interested
Edit	Copy	Delete	17	Pending	2025-07-06 12:18:13.856124	2	2	Hey! I'm a single tenant working remotely. I'm tid...

10. Authentication Profile Database : The image represents a database table used for managing user roles and contact verification in a property rental system. The table contains six fields: id, phone_num, role, otp, is_verified, and user_id.

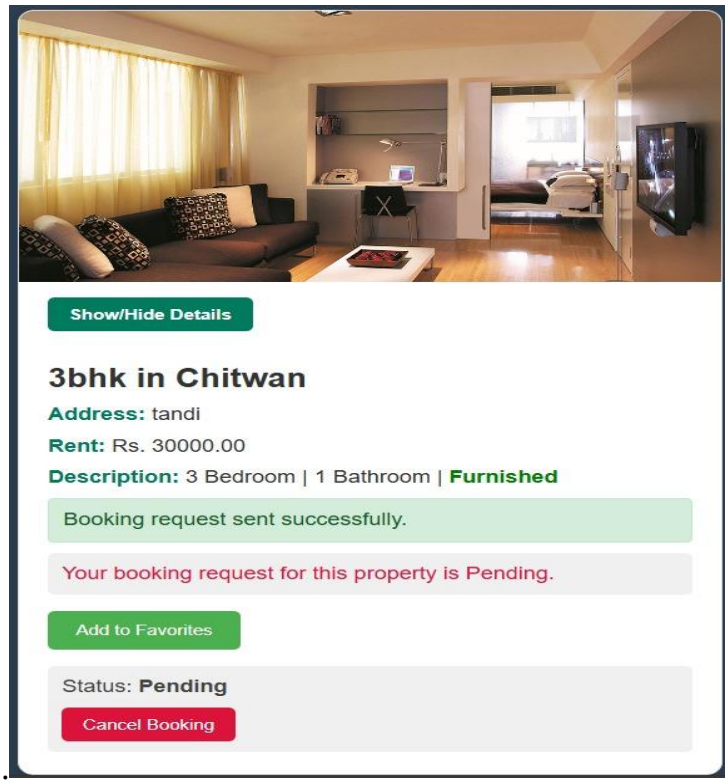
id	phone_num	role	otp	is_verified	user_id
1	9843961445	tenant	*NULL*	1	2
3	9855051790	landlord		1	5

11. Property Page(for landlords):

The figure shows a landlord's property listing form with fields for title, type, address, city, bedrooms, bathrooms, rent, availability date, image upload, and checkboxes for furnished and availability status. A green "Submit" button completes the form.



12. **Property Booking(for tenants):** The figure illustrates the property booking interface as presented to tenants. It includes a preview image of the rental unit along with key property details such as title, address, rent amount, and description (bedroom/bathroom count and furnishing status). A red "Cancel" button allows the tenant to revoke their booking request, while the current booking status is displayed as “Pending.” The layout is designed to provide a clear overview of booking details and enable basic interaction within the tenant dashboard.



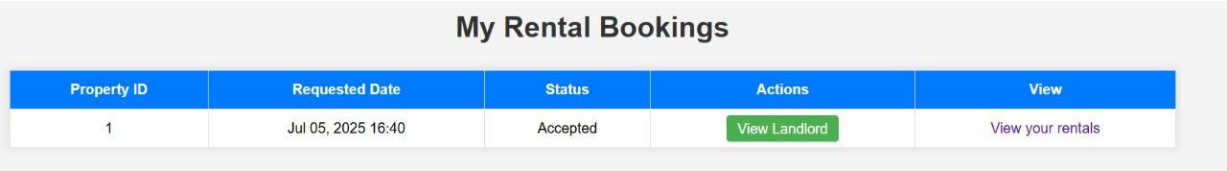
13. **Rental Requests :** The image shows a user interface table labeled "Rental Requests," commonly found in property management systems. The table includes columns for Tenant Name, Email, Status, Requested At, and Actions. Action options ("Approve" and "Reject") are provided as hyperlinks, enabling administrative decision-making within the system's workflow. The interface demonstrates real-time request handling in a web-based rental management application.

Rental Requests

Tenant Name	Email	Status	Requested At	Note	Actions	View
purnima1234	purnimapoudel46@gmail.com	Pending	Jul 06, 2025 12:18	Hey! I'm a single tenant w… Show more	Approve Reject	Property pid=2
ram1234	hopefulbrahmagupta@fearlessmails.com	Pending	Jul 06, 2025 13:00	Hi! I've recently started a … Show more	Approve Reject	Property pid=2

14. Rental Bookings:

The image presents a web-based user interface titled "my rental bookings," designed for tenants to view and manage their rental requests. The table includes columns for property id, requested date, status, and actions. The "cancel" action link allows the tenant to withdraw the request. This interface supports user-side functionality in a property rental system, enabling real-time tracking and modification of booking requests.



My Rental Bookings

Property ID	Requested Date	Status	Actions	View
1	Jul 05, 2025 16:40	Accepted	View Landlord	View your rentals

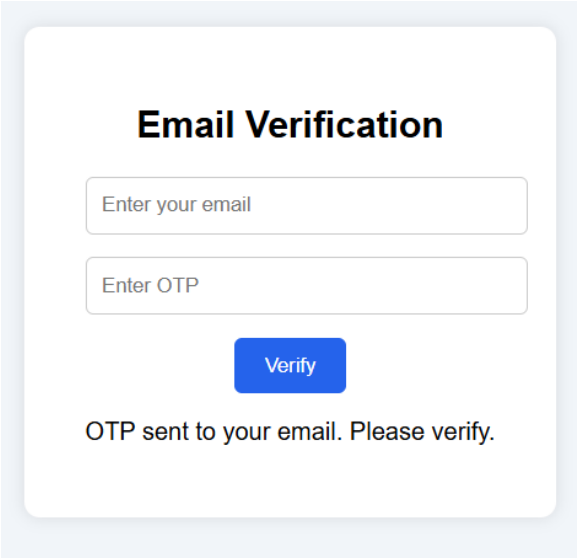
15. User Authorization Database :

The image displays a database table representing user account information within a web-based application. The table includes fields such as id, password, last_login, is_superuser, username, first_name, last_name, email, is_staff, is_active, and date_joined. The timestamps indicate user activity and account creation history. This schema is typically associated with authentication and user management systems in Django-based applications.

id	password	last_login	is_superuser	username	first_name	last_name	email	is_staff	is_active	date_joined
1	pbkdf2_sha256$6000	2025-07-03 07:15:09.753890	1	purni			purnima@gmail.com	1	1	2025-07-03 05:50:26.020163
2	pbkdf2_sha256$6000	2025-07-03 07:20:31.210193	0	purnima1234	Purnima	Poudel	purnimapoudel46@gmail.com	0	1	2025-07-03 05:53:39.401359
5	pbkdf2_sha256$6000	2025-07-03 07:28:26.221625	0	moon1234	Moon	Light	ml0961413@gmail.com	0	1	2025-07-03 07:27:42.597961

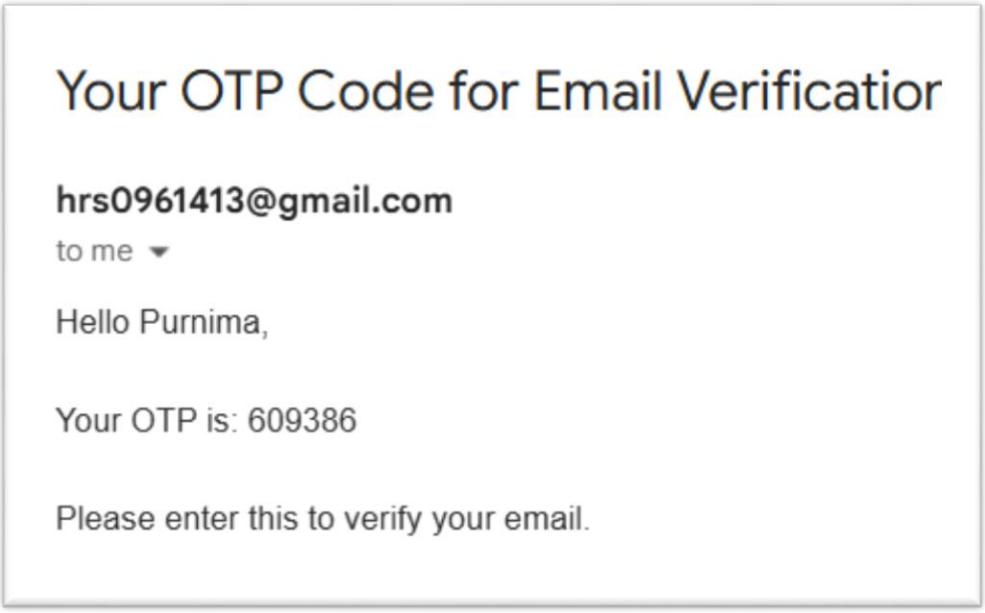
16. OTP Verification page :

The image displays a user interface component for email verification using a One-Time Password (OTP). It contains two input fields: one for entering the email address and another for the OTP. A "Verify" button submits the form for verification. Below the form, a message confirms that an OTP has been sent to the user's email. This form is typically part of an authentication workflow in web applications to ensure email ownership and enhance account security through two-step verification.

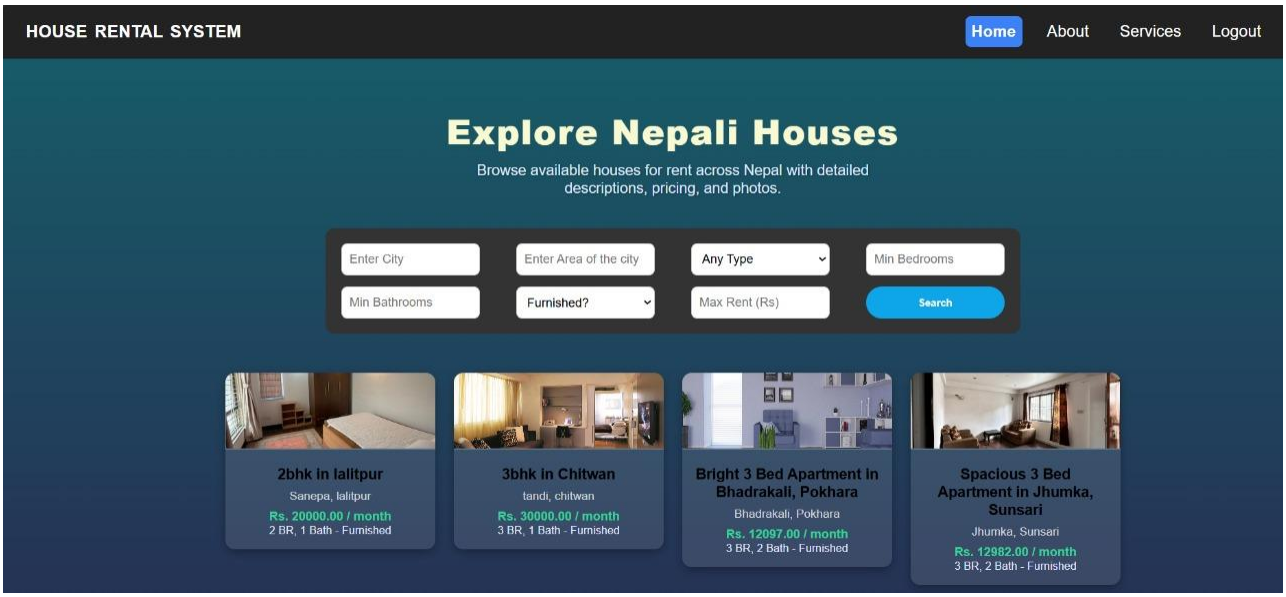


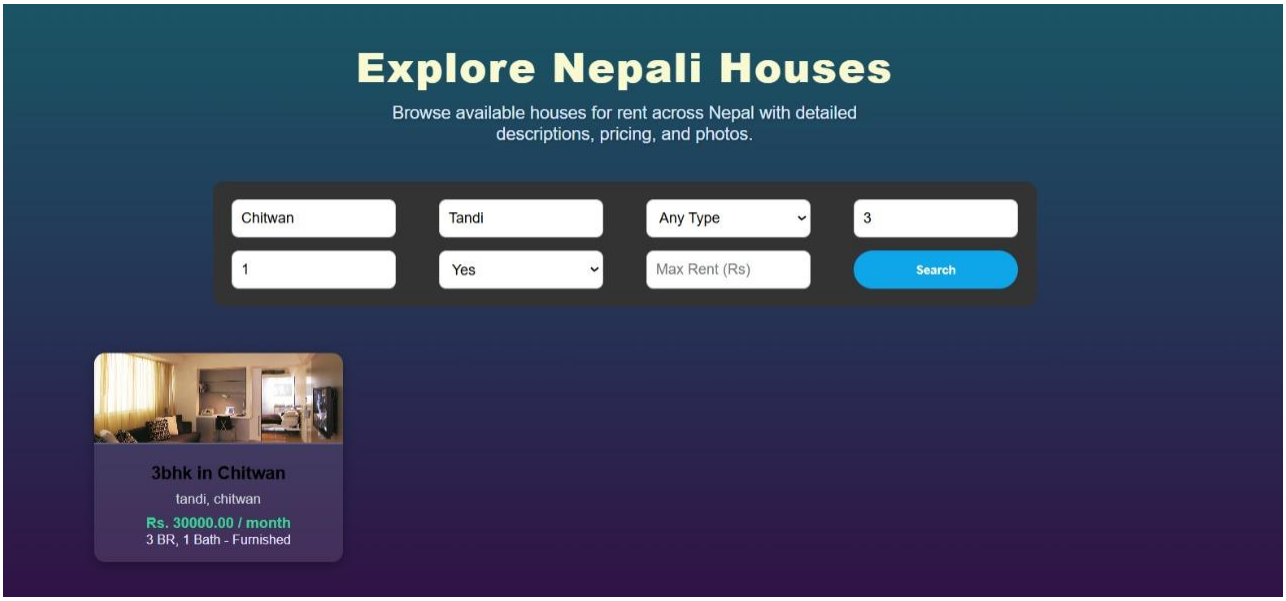
17. OTP Sent in Email :

The image shows an email-based OTP (One-Time Password) verification message. It is sent from hrs0961413@gmail.com to a user named Purnima. The email contains a six-digit OTP code (609386) and instructs the user to enter the code to verify their email address. This mechanism is part of a two-factor authentication (2FA) process, enhancing security by validating user identity through a temporary, time-sensitive code sent to their registered email.



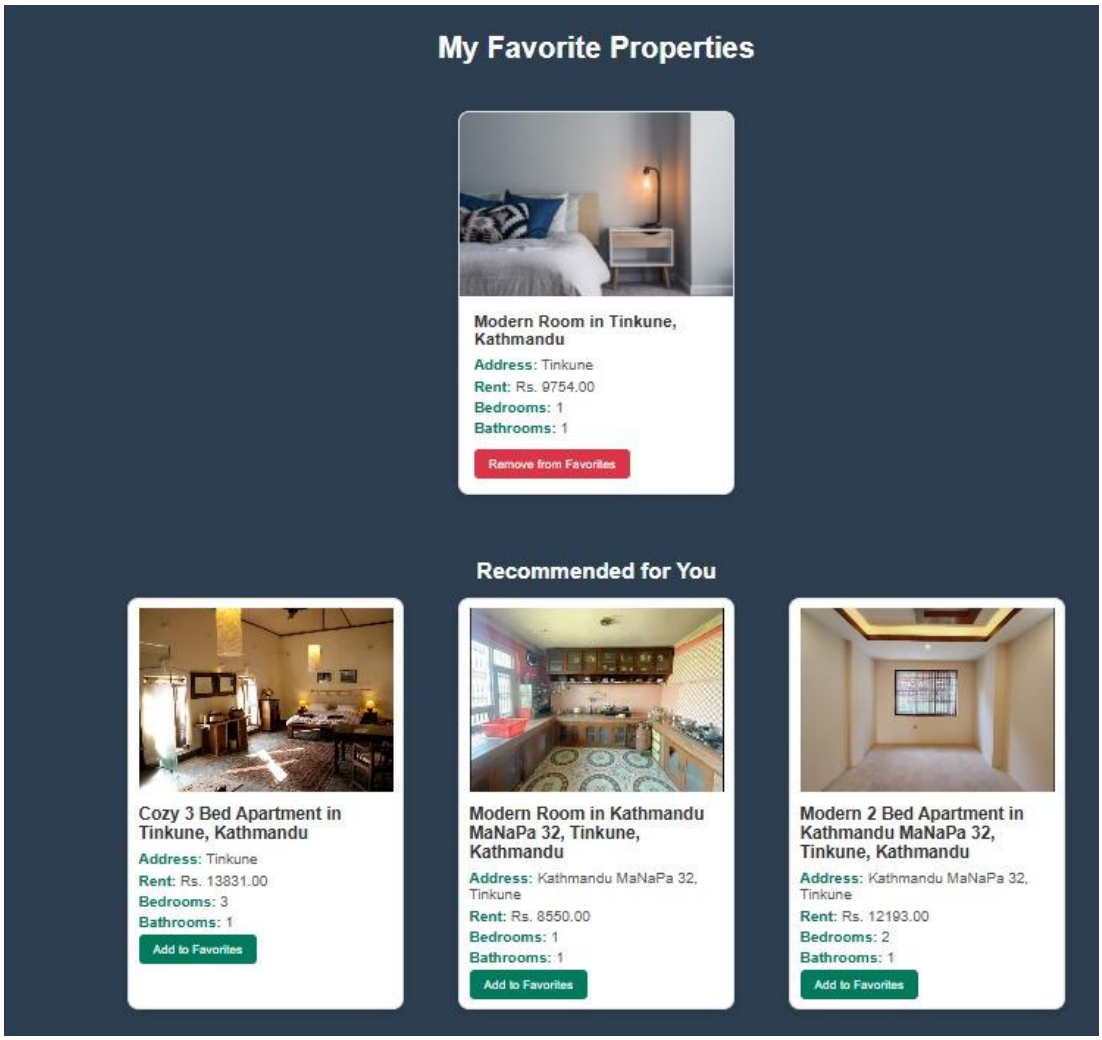
18. Filter in Data: The interface allows users to filter rental listings based on multiple parameters such as location, property type, number of bedrooms, bathrooms, furnishing status, and price. A search button triggers the property lookup. Below the filter section, search results are displayed as property cards with details including room type, location, and pricing.





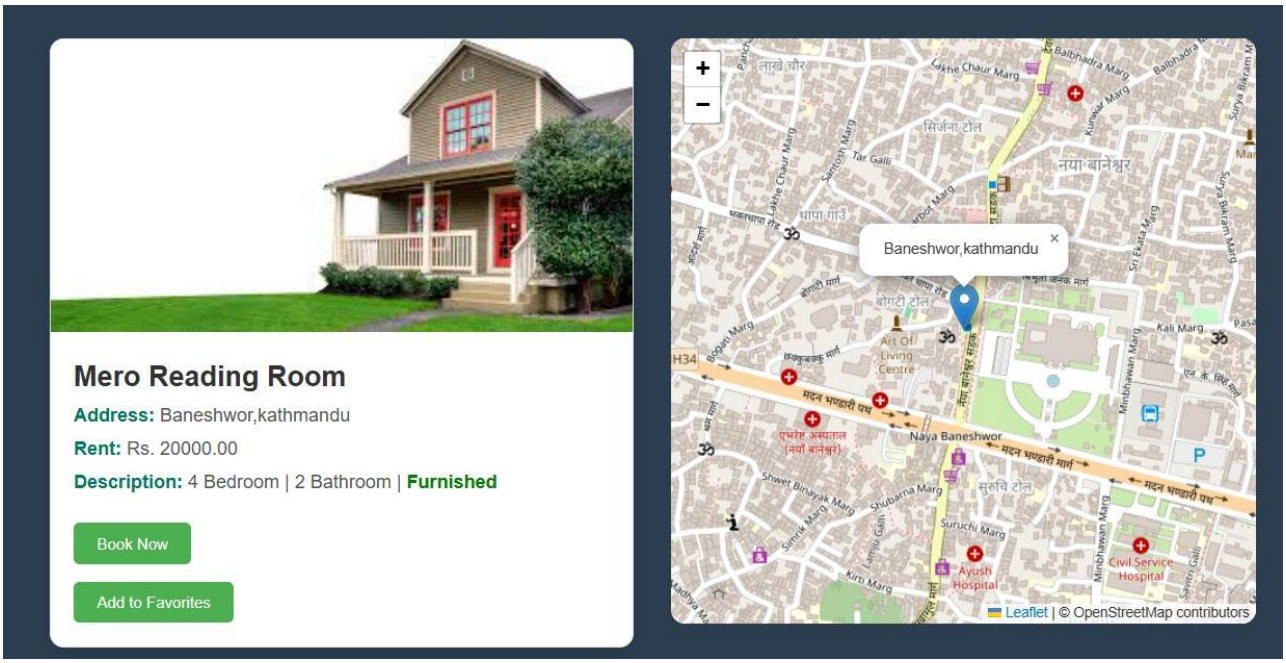
19. Recommendation in Data:

The interface displays a saved property at the top, showing detailed information such as room type, location, number of bedrooms and bathrooms, and rent. A "Remove from Favorites" button is included for user interaction. Below, a "Recommended for You" section presents multiple suggested properties in card layout, each containing minimal property details and an option to "View Full Details." This layout demonstrates the use of personalized content delivery and user interaction in rental recommendation systems.



20. Outstreet map:

The Outstreet Mapping System uses Leaflet and OpenStreetMap to display rental listings like "Mero Reading Room" on an interactive map. It shows real-time property data with markers based on address and includes details like rent, rooms, and furnishing. Features like "Book Now" and "Add to Favorites" enhance user experience, helping users make smart, location-based rental decisions.



CHAPTER 5: CONCLUSION

5.1 Conclusion:

In short, the House Rental System is an effective step towards automation and simplicity in renting houses. We have been able to design and implement a web-based system throughout the project that makes browsing, listing, and booking rental houses a matter of ease. The addition of search and filter, user authentication, and role-based access has been an important step towards the betterment of the rental process for the landlords and tenants. We would like to express our appreciation towards our project supervisor, faculty members, and all the team members who have contributed their effort and expertise throughout the development process.

5.2 Scope of Future Development:

Our House Rental System is very promising for future development and inclusion of more sophisticated features. The areas which have been identified for future improvement are as follows:

A. Mobile Application Development:

For increased use and access, the website can be converted into an Android and iOS mobile application. This would allow users to manage rental activities, search listings, and receive notifications using their phones.

B. Adding Real-Time Chat System:

The addition of an instant messaging system between tenants and landlords would ensure faster and clearer communication throughout the rental process.

C. AI-Powered Hybrid Recommendation System:

Although a content-based system has been suggested, the future direction involves the development of a hybrid recommendation model grounded both in user behavior and property attributes in order to make even more precise property suggestions.

D. Verification and Review System:

The addition of property verification and user review/rating systems will ensure trust, safety, and listing validity, minimizing the potential for fraud or false property information.

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